A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

John A. Janik
john.janik@gmail.com

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Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \ldots, b_{M-1})$, $B_N = \sum_{n<N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1}; "rep bits" is the bit length of its least positive representative r_N, "mod bits" is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1.

#	mean b	$\lvert D_3\rvert$	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997

#	mean b	$\|D_3\|$	$P(16)$	B_{240}	rep bits	mod bits	fill
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000

#	mean b	$\lvert D_3 \rvert$	$P(16)$	B_{240}	rep bits	mod bits	fill
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

3 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$\lvert D_3\rvert$	B_{240}	rep bits	mod bits	fill	rep bits at N=100/200
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

```
122121221212212121221212212121221212212121221212212122121212
212122121212212122121221221212212212212212212122121212212121
```

$\theta = 0.10$:

```
122121212212122121221212122122121212212122121212212121221212
122121221212212121221212212121221212212121221212212122121212
```

$\theta = 0.20$:

```
121221212212121221212212122121212212212122121221212121221212
122121212212122121221212122122121212212122121212212121221212
```

$\theta = 0.33$:

```
121221212122121221212212122121221212212121212212221212212212
121221212212121221212212121221212212121212212121221212121212
```

4 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_N+1} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x+1)/2^{v_2(3x+1)}$ and a period-p word w with sum B satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \qquad A_w = \sum_{j=0}^{p-1} 3^{\,p-1-j}\, 2^{B_j}, \quad B_j = b_1 + \cdots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

$x = -1$: word (1), period 1

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \qquad v_2 = 1, \qquad S(-1) = \tfrac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every k.

$x = -5$: word $(1, 2)$, period 2

$$
\begin{aligned}
&3(-5) + 1 = -14 = -2^1 \cdot 7, && v_2 = 1, && S(-5) = -7,\\
&3(-7) + 1 = -20 = -2^2 \cdot 5, && v_2 = 2, && S(-7) = -5.
\end{aligned}
$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \qquad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

$x = -17$: word $(1, 1, 1, 2, 1, 1, 4)$, period 7

$$
\begin{aligned}
&3(-17) + 1 = -50 = -2^1 \cdot 25, && S\colon -17 \mapsto -25,\\
&3(-25) + 1 = -74 = -2^1 \cdot 37, && -25 \mapsto -37,\\
&3(-37) + 1 = -110 = -2^1 \cdot 55, && -37 \mapsto -55,\\
&3(-55) + 1 = -164 = -2^2 \cdot 41, && -55 \mapsto -41,\\
&3(-41) + 1 = -122 = -2^1 \cdot 61, && -41 \mapsto -61,\\
&3(-61) + 1 = -182 = -2^1 \cdot 91, && -61 \mapsto -91,\\
&3(-91) + 1 = -272 = -2^4 \cdot 17, && -91 \mapsto -17.
\end{aligned}
$$

Cycle formula: $p = 7$, $B = 1{+}1{+}1{+}2{+}1{+}1{+}4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \ldots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$A_w = 3^6{\cdot}1 + 3^5{\cdot}2 + 3^4{\cdot}4 + 3^3{\cdot}8 + 3^2{\cdot}32 + 3{\cdot}64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \qquad (139 \cdot 17 = 2363).$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the -17 cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \leq C p^{\tau+1}$) governs.

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|$=0.214):
111411211121121111112114421114112222111122212112122212112221
112221122411212122222121212111211111121221222111411112211121
222111111112422222211111121211411212111214221114142121111141
211211412222121111111122222121141212112214111211111114422241

#2 (mean 1.579, $|D_3|$=0.214):
222222222222222111111211211421111211421111221222222111424121 1
111112412111211422111141112111122112212221211111112121111211
212222111212111114221424111211121411211144212111114121114221
211211111111122121111121211221121211141121141242222122222111

#3 (mean 1.579, $|D_3|$=0.214):
222112111112112421221112111211112121221112111121121221144412
422222211122121211111112111111121141212111222422214111212111
112411112112224211112111211122121122111124112111224111121221
224122114112111122122412112421122212111441121111211212111122

#4 (mean 1.583, $|D_3|$=0.239):
111111111111111111111111111111111111111121222114114214122111 4212
111421141112214212421122114411224111211122121221122222221111
111112221221121111111111121421121124112212121142244122121211
121411412211121111211221111422424421212112121142112221121111

#5 (mean 1.592, $|D_3|$=0.220):
111111111111111111111111111111121121111412414211121212411112242

12112112112121111421114122211124212211121122211111111122111
221222112412112111422412112222244212121411111124111221221212
2122211211411111122112142111122211122224114212211221214211224

#6 (mean 1.579, $|D_3|$=0.215):
2222222242222222421121111141242121141122122222111112112241221
111121121111111111111221111122222421114212214211111112422221
111112221111112111222214111122221121211242111111411121211222
111112111112111112211221214211141122121141214111121211212112

#7 (mean 1.592, $|D_3|$=0.215):
121112111211212121111224121112121411212114112411221242222211
221221111112211111221212224214111211122112111222111211112112
212422141111221412421111222214112142112111121212111112211111
111141212222242111111221211221111212421212114112211222141211

#8 (mean 1.579, $|D_3|$=0.216):
12211111111112121121111141121222211111211112111111211142211414
222221141211111421222112121111112114122121211122121211421121
41212112411221111242211222111111411421211122222411112111221
214122121114112111211221141224111112112112412121221211222221

#9 (mean 1.579, $|D_3|$=0.229):
2222222222222242222222222222222222222224222111222211221144111
1211211221112211122111111211142124111122211121121111122241111
24121212114111112111211111212211111221112111114121212114211111
1111221122111111211111142242121121121111221111111111112112421

#10 (mean 1.579, $|D_3|$=0.233):
1111121112221212221212111141111414222441214211212124111122211
111111111121121421111411122221211224211422141122144142111121
14214111121221111212222411111111111111122422121212211111111
111221422111212111111111142211121111212111211211121114122221

#11 (mean 1.579, $|D_3|$=0.229):
222222422222242222222112211221241221221112211142211112122112
1111121112112121111211122112112112211221111222241121211241141
121211114211121121122111412121211214222111112112411211222121
224221212212212112111421211121412211111112211111111212211111

#12 (mean 1.579, $|D_3|$=0.216):
422211212114121122244111122221112411221221112121112441124421
1122221111121112111122121112212212212111111111121242212112122
114122214121111111241241112114112121111111411122221141211212
21111211121111111112224211211142111112122121211211111212211221

#13 (mean 1.592, $|D_3|$=0.233):
1111111111111111111111111111111111111111111111111222212111121
122124214112211111112211111222122141114421122211212442114111
121241421122224122111422121211222421122114421142422121221111
211124144212122112121212221112111241112111121211114212122141

#14 (mean 1.579, $|D_3|$=0.215):
24222222224222222211111111211221111111111111111111221121224211
211121212222112122121412112122112122121121212111121142111121
212111221114111121121244122211211122111122124124112112222212
211211111111211214114111112124111224141124122121211111211411

#15 (mean 1.592, $|D_3|$=0.216):
111111111111111111111111111111111112441112211121111112112211221
121221121141241214211241221111111224121122222211111222211122
122212222142121212142222241121411121121112221411414111122111
211211221121242112114124211211224114212222211141122222111111

#16 (mean 1.583, $|D_3|$=0.214):
2122141111121111212111112122111212111121122414224112111221122
2411221421212111111241211244411221221111111222212124222211214
211121421111112122111124211221211412111111221122111111121222
1141212114112111111241222121221211112121212212121121111211122

#17 (mean 1.592, $|D_3|$=0.215):
111211212242121222211221222111112122421112112112222214111212
211111242122412112112121112211141122141121124221112221211411
411111421211221112112211112211121111111221221121221121114112
2122211111111141114112112141111112221121242111211222242421222

#18 (mean 1.579, $|D_3|$=0.218):
222222224242211124111122122121141122122112121221122114241112
111224211221112121114111121211411111222121121212412111121222
121141211112222211121111122114124141122111221111212212221111
212214221121121111212121111112221114211211122111121111122111

#19 (mean 1.592, $|D_3|$=0.220):
112411111211112121112411212111212211211111412211111211112221
121212111212212112121114111111121411221224421211221122221112
211414211141112112212244111111211211112112221412411141212222
111121122124411212111122121241121241121214242212112211111111

#20 (mean 1.579, $|D_3|$=0.260):
222222422222222222222222222222222222121221111122111111221124112
211112112211224121114111122211212212112221111111412122212212
421114212124221212112112211111121122111112211122124221111211
121211111121222121111211111111111141121112411212221222121111

#21 (mean 1.579, $|D_3|$=0.272):
222222222222222222222222222222222221212111122122111211222222121
1111112212112111112221111211112111112221112122111141121414121
111211211121422211211212121122421212211112414212111121121121
121121112111111122121211221121122212222124412212111111412122

#22 (mean 1.592, $|D_3|$=0.220):
111111111111111111111111111111211112242124211142124122111112241
221114214212211122221221414421121112221121141222111211112212
114111112111411212111121111224411212111221121222421112111114
121111121211212124242212121241212211122411111212122122211212

#23 (mean 1.587, $|D_3|$=0.228):
14222211241141111411121111121121122414422122111111211211222412
222121111211221211111211122114112121214111111211111221122122
124221212111111412142112112221111212111114121112421211112112
1224111111111111111121111222221111224244111241142141111122111

#24 (mean 1.587, $|D_3|$=0.246):
111111111111111111111112114212211111111111111111111122122211412
121212122414211411111111214212211411121124212122222414114212
211414122212111221211121112111241111111221142111211112122141
212111142121111214124412121221222112221411121421141114211112

#25 (mean 1.592, $|D_3|$=0.222):
111111111111111111111111111111112422122111122222112242111111411
1211112221221112421142211241212122114111121122124112221241421
221112412222221212211122224224211212124211214222221211111122
211111121211221122122121112242221211122221111111122122112211

#26 (mean 1.592, $|D_3|$=0.242):
111111111111111111111111111111111111111111111111111111121111111
221222221111122142241112242111114111111221112144222421121124
214244121121122112141121212241411421221111411122211122141211
112112222442112212212412211111114124222122121141242212211211

#27 (mean 1.579, $|D_3|$=0.222):
224222222222222222222222111122211112121112111121111111121111111
122211221242222122121111111112214111212221211111421112141422
411121121111111121211222411411221122211111111122111212112112
221424122222421221111111111411112142222111111211422212121121

#28 (mean 1.592, $|D_3|$=0.214):
111111111111111111111111111111112111441121411121211121111121122
4112141121112221221122122122222224221224121112122212112412111
112112121421211211141112211124211241122421222221122112122121
121211122211112112144241211122442221121211112212111412211212

#29 (mean 1.587, $|D_3|$=0.215):
112112414211111111212422111111111112111222111411211244211212
412112121222112122221211221211112211114211221222141212121124
111211112422112111242412111211122211111222121112212111121112
112212121211111442114414211211212112114211221212121222111111

#30 (mean 1.579, $|D_3|$=0.260):
222222222222222224222222222222222222222222214111121121111112211
221111111211111241112221211224142122121221141121211212121122
111111112222111211121111111111121111111114422212121214141221
212211211211122111111112121211121122224122121111122211222211

#31 (mean 1.592, $|D_3|$=0.214):
111111111111111111111111112112211221112211421112212222222221214
212222111112121211122212112212222211111112212214124121121111
141121112111141114211442224211222241211111412111211441122221
112221421211111121112211242111224111122122412211222121421111

#32 (mean 1.583, $|D_3|$=0.239):
1111111111111111111111111111111111111111112111124112211421214211
241122242112114144211221211112111221121211121222112212122112
112211112111114121142142122114224221211214111221212111412212
2211111121412211121114114441114212212411221111211112221224111

#33 (mean 1.579, $|D_3|$=0.218):
421241222121121121112121121222111121241111111122212112112241
222111121111112141221124222122112111211411111222112211121111
1111112122111111221112211222221244111141242121122112214221112
121221211222212111221112211114212222211112422211111111124412

#34 (mean 1.587, $|D_3|$=0.217):
1141112111111111411112144142414111211211122111122211122111122
112111222222211221122121212211221122111111112111124211221211

1124121221212222241224212111111112141112212422112112122211412
2112212112121112111411111111111114114214112142222111121421121

#35 (mean 1.592, $|D_3|$=0.214):
1111111111111111141222111211121214214421221212422121212412l4
1121121224211221111214112412222122421112211212222411112121 22
1111111212112121412122212112412412222111111411221112122241221
2221111221111112111211121122412112221111121241111211112111111

#36 (mean 1.592, $|D_3|$=0.216):
1111111111111212122221212111112121221142214112122222121121211
1121121122121111112114121142121112211211411111141111121124211
1111421122114221221141114112242111121241412214222211242 14121
2124111111221121211112114111121221221222111224112122111212211

#37 (mean 1.579, $|D_3|$=0.220):
2222222411111111121211244221212111111211111111141212124111224
1242111112111221212121114221111112121111111111221124141122 12
1122112142111212242111221211111111112124124111111424112111 11
2221211122421122412211112222124111121211122412112211112111 21

#38 (mean 1.592, $|D_3|$=0.225):
1111111111111124211211212112221221211112411212111121111241 1112
1212221224422112221142422112111112222121141111211221211122142
1412144421111241242241211212111214212141121111221111111241121
2211111411121111211121111211111112211221142221111111121214112

#39 (mean 1.587, $|D_3|$=0.221):
2241112122211411111122412221114112111111111411212111241211211
1112111222212141111422114112211221122121112122121111121111111
2414112111212421212111422221111141211211211111212121121111111
1442212111121241221221112221112112111422121211112244421 12211

#40 (mean 1.579, $|D_3|$=0.218):
2222222222221141122111111214111211111211111211121111222411 22111
1121214111112241112111122221221122111221121122121114121422 12
2221221222212221121111211141122211212111121411112211222114 11
1211112112124114122122122121212114112214412111111111242111212

#41 (mean 1.592, $|D_3|$=0.225):
1111111111111111111111111111111111111221121111112214122141 2211
2111122211121412211221111222111111122111211214212414242412 11
4221211111411111141111241112121212221112221222141222212212 12
2421212241114214422121111122211121241111412122421111124122 12

#42 (mean 1.579, $|D_3|$=0.238):
2222222222222222222222222222221112211141121111111114112412 1111
1212411124111221121211111112111211211111222111121122121121 21
2111141211111122111122122121212111212111211212111111121221 111
2122411124111221222122111122111242221141121212122422222124 41

#43 (mean 1.579, $|D_3|$=0.215):
2224222422211212242112141412112111244112121111121111124111 222
1111122211112224122111111111121212114211211112121122242142 21
2211112121111221221121211112111112122111122212211211111112 2214
2122412112111112212121111211211422111111142111112122214121 21

#44 (mean 1.592, $|D_3|$=0.215):
1111111111111111121111221221122111114211122111112242111212 11411
4222111122111112224421111122111121224121111212221412211422 211
1122142112211111211111221224112121122122112112212111121111 121214
1121121111214111112222114122221121212221224411442221222112 112

#45 (mean 1.587, $|D_3|$=0.217):
1211211224212211141411111111112121224112211111111111211111 2111
1221122222112112122241112411412222412421412112211212214214 12
1111111222122121114221111212211121121141212111122222222211 412
1111111122121211211111212141112411111411212221121422111111 41

#46 (mean 1.579, $|D_3|$=0.215):
2222422222222222222222222121111121111211214221114411221111 112
1241111111121211111211122221111121221211211111221111224122 114
2111211444111114212211122221111121111121221211211122222111 11
2141121412412141121142111121212112111211421121111121111121 212

#47 (mean 1.579, $|D_3|$=0.229):
2242222222222222222222242224422222221221111241111441142111 1111
1111212112111112414121111121221212111122211212112112221112 11
1114111212111221111112212122221122112121121111111111212211 111
1214211211222122421112222122221212211111111111211121211214 11

#48 (mean 1.579, $|D_3|$=0.215):
2422222224212211121111111111212111244114111111112242212142 1112
2111212222122212411121122122122121121212212124211111411112 11
1214141111111111121114112112122221421211121221121222121112 21
2424212212111111111111112222111122211211114112111111211221 2112

#49 (mean 1.592, $|D_3|$=0.215):
1111111111411211422221221144112112214212142421211212442111111
2122211211211212214212211411212111212212121122222212121112111
2211111211221111111121121241112111221111122121211121211111122
2222111121141212222114421421222111112411112222412121211111111

#50 (mean 1.592, $|D_3|$=0.216):
1221121214211112421211221121212112212122214214211111211111121
1412211111414211112221221214121221111144112111111111111121221
2111441211121112112122112211222212114121211111111211141111212
1214112421242122121221221241122111141211121121224211111221211

#51 (mean 1.579, $|D_3|$=0.229):
2222222221221242111224122211221121121112211222141221111112112
2111244112111214211121114111122111112224421111111121122211121
2121221211111222222111221211412212221121221112112111122112221
1111121222211411141141121211111214112221211212211221111212112

#52 (mean 1.579, $|D_3|$=0.248):
2224222222222222222222222222222222222222212211121111221111112121
2111222111111121111241111122222112111111112222111121121122111211
2421111111221121224212221124421124141211112211122111121111111
4111122122111212111111241122112112121144112111211211211221121

#53 (mean 1.592, $|D_3|$=0.218):
1111111111111111114212111442211121111212112114212214111111121111
4121221214222222221211112121121222222141111221212111411222212
2144111142411111214221112212121111212111212121241111222111212
1112122121122211212242222121221141111211122122112221222111211

#54 (mean 1.579, $|D_3|$=0.215):
2422222224422222224422222222222421111121222111111121111111111142
2112111121222211121211111421122211411111211221221121111122111
2112421112212111111212111421211111412121211211111211111221122
1111442212212111422111121121111222111111221121111412111111211421

#55 (mean 1.592, $|D_3|$=0.233):
1111111111111112114411114141214221121211122111121221112111111
2211212111112221211111122112112142111224121111222122111141112
2111121421121212411122142241124112212412111121121121111212112
2211411112422412121412111111411211424222121122114114111121121

#56 (mean 1.592, $|D_3|$=0.279):
1111111111111111111111111111111111111111111111111411124212411 2
2411221111111211211421111141212242211111221212442111121111141
4421222214111211211221121421124224111142112142112124111112411
2111111222121221221412111112212144122112122114414111111211142

#57 (mean 1.579, $|D_3|$=0.215):
2222222222112141122112112211111112111411222122242112111121111
1112114211112222211212412221221411211212411111121111211241222
2141221121121411111111121421124211111221214111212222141211211
1221121111221422121111114222212421122112112111211112121111111111

#58 (mean 1.579, $|D_3|$=0.218):
2222222222222111112121114441211141211221212111112112211241244
1221114111121121112112111111112122212142122111122121111441112
2211211122111242211211224122212111111111421211124211124121112
1211211221211112111122121211222112211122121111212112211222211111

#59 (mean 1.579, $|D_3|$=0.238):
2222222222222222222222222222222222222222242222211221111111122211
1121121221121121444112112111141212411121122121111211212121111
1124211142114222112111221221111121124111212114121111111121212
1111121221111112111114211112211121112111111142112111211121211

#60 (mean 1.579, $|D_3|$=0.214):
2224222222222222211221112421112111241212211142111211211141221
1111121121214112211111111224121111421214114111112211111141112
1112112122114141121114412212121112121122111242122211111122221
1112421121121111121211112114121111112211111222112212222221211

#61 (mean 1.579, $|D_3|$=0.238):
2222222222222222422111111111211411112121122211142421211111111
1211221121242114142211112222211122111111111212211111111212112221
2122214212111122122211221221111211111111112214221111221111122222
2121222212112111211422122121441221112112111221114211212121112

#62 (mean 1.583, $|D_3|$=0.230):
1111111111111111111111111111112412211111211421211224214141121 2
1112212221221411111211114222114121141114112121111121211212241
1122114124111121112242114121211212122244122421111141212111411
1221221111122112411211211111221122221211212112122122211114122

#63 (mean 1.579, $|D_3|$=0.222):
2222222222222222222222211412141121122211221111211111224111122 11
1112211111212122141422111221114112112411121122111112142112211

21211421411111112121111211111411142142222211121111212212411111
22212241111211211221111111111221211121112221221122221211122111

#64 (mean 1.579, $|D_3|$=0.215):
22222222222222222222221111114111121412114111112121224121111 2
21112112111121421111114122212212111114411111111411211124211 2
21111141221122114112221112112141114211211211221111122121 2211
22121211211211112111111411242211221121221112142211121222 1211

#65 (mean 1.579, $|D_3|$=0.222):
22221222112121111114121121411112222121111112121121221221222 11
11124241124212111221112121111111221112112122222221121111212211
11211212211221221221111122112122221411211121211412212121 1211
14212142111411221222222212111241121121111414221121111 1441221

#66 (mean 1.592, $|D_3|$=0.216):
11111111111111111111111111111111111111111111111121212112211111 1
22112221241224211221111211121111122122241114222121211222421 1
12411241121144112112111222421412114111122112222221212121212 2
22221121211424121244242112111212142121112421124222121211221 1

#67 (mean 1.592, $|D_3|$=0.215):
11111211211114111111222212121114114111211211111212124114122 21
12222211121221411411221212222121121141121211241121111221122 12
21411121411122122111214111211111212112112212241214111222111 21
41211114421121211121241221121212221211121212221122211121122 2

#68 (mean 1.579, $|D_3|$=0.216):
22222222222222422424241112121111212412211442121121141111112 21
21211212111111244212112212222211212111211121211211141211111 2
11121211141221211114121112111111122122111221111111211241121 2
12212111111411111114411221111111122421211212411122211111211 1

#69 (mean 1.592, $|D_3|$=0.216):
11111112122411111221211114111112212112122241122212222221142111
11111422121124221214112222424221121424211122111122121122211 1
14111111212121221122111112122111122114111111242212111211111 1
12124112221121411111141112242112111211141212112112111142121 2

#70 (mean 1.592, $|D_3|$=0.216):
21114212112241222411122411211211212112112121121111114121122 22
11122111212141212212121111111224112121121212111112211121422 4
11222112121121212114221122211121121141114222111222411111111 1
12412212111412111112212122112111214421111114422412111111111 2

#71 (mean 1.592, $|D_3|$=0.214):
11111111111111111112211214121421112122112211114122144221122222
11441441242421122214111211212121111112111421222111222121221 1
12222421122121112122221241111141121122111211211111212111112 22
21121221121421212111121111221121241211211122141212111211111 12

#72 (mean 1.579, $|D_3|$=0.214):
24224222221221111141112111124112112112121221122111122121212 1
22441111422112211122411111121212111121121211421212114124211 1
21211121122211222141224111121111121221422121211212211112212 1
12221221212121121111121112141121111211122411111111211411111 4

#73 (mean 1.587, $|D_3|$=0.221):
12114111112241112112112422112121221141122211112111111242142 11
12222111111112212111212141211144111221211112121111122121212 1
11221111111241242121122212211111412112411111111112111211212 41
21112224221411112241121121112122214121212224111122111114112 41

#74 (mean 1.592, $|D_3|$=0.215):
11111111111111111111212211212142222211411112212121121211221 11
11122122114124211222211112212212121121222412112222114221221 2
21222122111212111111222421221111111121212121211111211424414 2
22141121112411121141111211111111221111412242212222114112111 24

#75 (mean 1.579, $|D_3|$=0.215):
22112411142121212121211122221111111242121121212112114122121 1
21211141211411112121111211411212412112411112112122122122441 2
22111122221111212212211111121212122222114111211142222111121 1
11421122112112121221211121221241241121211112122112111111412 1

#76 (mean 1.579, $|D_3|$=0.225):
22222421112214111221111124211111112111211212411111121411111 2
42111112111121212111112241211121421212111211111111222112212 2
21112212122121114112111242212421112411221121111111111211241 4
21114121211211111211112122212141422111121122111441221442111 2

#77 (mean 1.583, $|D_3|$=0.241):
11111111222211121111121121111222121141412224214222222124214 2
22112211412422121212211211112412111212121112112111111212222 21
12112211212114222211142222111124111121121112211111122122112 2
12122122112421211121211221221211122122111221211111221212121 12

#78 (mean 1.592, $|D_3|$=0.215):
1111111111121121112122211442211222212222221111121122212111412
1122211222121142414111122212111121114111212441111121122222111
2212222121121211121122214412211111111412122221111121222212222
1111224121112211121111211211114221222142141111221111114411211

#79 (mean 1.587, $|D_3|$=0.221):
1412112211111114212212121112121122221112111111111411111111124
1121141422212211111211221421222212111211221111111144414124222
1111112411214111211122111111242114121111421112212111221121112
1241121411122211122121221121112411411121212212211211111222242

#80 (mean 1.587, $|D_3|$=0.217):
1111121442111122111412121121112112221111221212111212211111121
2112211111214142122121114221422121112112111211221121211122142
1211122122121222211412222121211121111111221241111114211122242
1221122422211124111421212121121124242121211122111211111121112

#81 (mean 1.592, $|D_3|$=0.225):
1111111111111111111112414111122221121412121211114121142114 21
1122121111111211211212412111112242411111112211421121111212141
2222112111211422111222411124222111111141221121221241222111112
1111121211122212112241142111242212421222142111111121412211121

#82 (mean 1.583, $|D_3|$=0.214):
2121121242412111121411212112122112112121121242211111111121214
2241112111211244122211221221112221121211221411422141124244 21
1242112211121111112112224111121121212111221112111111211221 2121
1111111112212211121411221112111122211111221122412112212211 11

#83 (mean 1.592, $|D_3|$=0.215):
1122121111242121122112111112112111114112212121222111111111122
2211112241112222122212221211112122221121121114224214412212 12
2411122212212111214141224422211122411112112121122111121114 21
4111121211211211414111112212122141121121211111122211121211 11

#84 (mean 1.592, $|D_3|$=0.218):
1221111111211411211222121422211422121122211112121221212112 11
1212111121121222421411111141121242211112122222111121114411 2242
1211111111122121122122222122114412211111212221221114111112 14
2111212142211421112121112111211411211112222121122222221112 221

#85 (mean 1.579, $|D_3|$=0.222):
2121224411222211422212212121121111121111211141411412111111 22
1221211111211212212112112112122112114122212112111211222212 11
2221242122112111211211241212141111221241111221122121111122 12
4412212212121111221212241112121212212122211121141212121111 11

#86 (mean 1.579, $|D_3|$=0.238):
2222222222422224222222222222211221111212241212211121242112 214
1121111111111111211221421111112211121111112121121412221111 142
2112211111122111121112121112141221212121121111221212211221 12
2121222211114114242122112121121111122222121211211141111121 12

#87 (mean 1.579, $|D_3|$=0.229):
4222222222222222242224222214222211222121211121121112211141 2
1112211121111114221112211111121124112111122122411111111211 11
1121222111111111111114111122411121122121111111221121124222 2
2211222211111211121124142112122112212224221111212121221122 1

#88 (mean 1.587, $|D_3|$=0.227):
2421412121211212121412111411122212121111121221121212121222 22
1211122211112111111212412111211122122111111112111121121221 21
1212421224122121112121222111111221212222112124212121212211 22
1122242111122221212442221111121414121111112114221111111241 22

#89 (mean 1.579, $|D_3|$=0.218):
4222222222114111211221112121211221111111212111111111122122 121
1211222112121221222111124121121114112114211112112121111211 14
2412211111221114111112241411212111211112422212222222121222 42
2112211112112112211111112221121111212241114214111241222211 12

#90 (mean 1.592, $|D_3|$=0.216):
1111111121121211142122121211121221122111212111112142211142 111
2111221111244121112222142121111111121211211211422111124211 12
1111212122122222121224142112211121212121212121112111411421 211
1111211411122121211112222212214114111122441421121112412241 11

#91 (mean 1.579, $|D_3|$=0.216):
4222224222212141112121121212141112111141112121112111112111 12
1121121212212411211121412212121111111111121122241112111222 12
1244122141421144121211211111124122111211111222212212121112 11
2212211111124122111111112111221121111122121141211142141212 22

#92 (mean 1.583, $|D_3|$=0.225):
1111112122221112212211411211221122421222211111242212111121 12
1241112224112112411211221121111111121224122422111111111122 12

2111211122121211121121111221211121212224222222222211122121212
1121121122212412424211412111121111112211111212421112412112 21

#93 (mean 1.592, $|D_3|$=0.220):
1111111111111111111112212421211111141412122212111121122122111
1122141111212221111112221241211211112412112121212212122112122
4424211221114211241121211112211211121121141112441141111124122
2211121112112221221212114242122421214112121114111221121221 11

#94 (mean 1.579, $|D_3|$=0.214):
4422222222222224224121111111111121112211112112414211221121 1
1121121221212242121224112111122111221242121121114111211112 11
1412421121111121221111222411412111112412211111111111142111112
2221112211111212111214211121222122111121121111214122121122 11

#95 (mean 1.579, $|D_3|$=0.238):
2222222222222422222224222222222222222222112111111111112212121
2411111121111121121221112411211211111214212112112121112112 21
1112412222111114111221221111114222412112111121211211111111112
2111122122211114111221222111124222411111111121111222412211111

#96 (mean 1.579, $|D_3|$=0.238):
2222222242222222222222121122421111211111212112124121211122122
2211222212111121114122111111112141122111221121212211111221112
1211141221141111221121122111122221121241412112121242212111 41
1222121111111211241221111122221211211111111221121112421122121

#97 (mean 1.592, $|D_3|$=0.233):
1111111111111111111121141422211112112211122221111121111142412
2112114214211142121212241114221241112124211142211222122111 12
1122112111111121121212221112111141212112222141214111112112 21
1214111112221411111122211221111411211412112411212244111124 11

#98 (mean 1.579, $|D_3|$=0.229):
2222222222222222222222211114121111112211122122211111112121221
1212111422112111221111412221221111221112111111211121211111122
1112122414111111114122222121111121111112121121122221141111 24
1241212224112112111124224122121142112111221112241121112122 12

#99 (mean 1.592, $|D_3|$=0.218):
1111111111111111114112211414242242111112412411112211211112212
1221111414112111221111241221211411211111221121244222212242 21
1222121222121211111112112222242141222112211122211221111212212
1221112111121222212122221112211112111222214111111211212122 11

#100 (mean 1.579, $|D_3|$=0.215):
2222222111144112112422112111412122112221214111212111111121221
2141122121222122212222211121214211221221112441141121111 11
1144111141121114114111111421122112211211211121212121212122 11
2121121111111212122121111112221122212212211112111211214112 11

#101 (mean 1.579, $|D_3|$=0.218):
2222222222222111121221211112112111421211212111111221141241211
1211141111121121221111111212124112222112221121211112111221 11
1422221121121121114111211121212421222222111142422212211421221
2224111211111412141212121111211112111121212112112111424122111

#102 (mean 1.579, $|D_3|$=0.218):
2222222222222222222222222422222222222121211224211222122412112
2121211112112121112212211411211111411242111111211111211141 11
1211221111121111111211111421121114211111111111122411124442 1121
1122111111112111211214111112111124241122111111221121211211 21

#103 (mean 1.579, $|D_3|$=0.229):
2222222222121111121112111224112222211212111221221112211111122
1221111212421121121212111212412121211212224421211121214422 11
2112121211111112422121121121211242211112211111222111121211 12
1214121112112121221122211124122212412111421111111212411211 12

#104 (mean 1.587, $|D_3|$=0.217):
1211422221212121221211142111221111221221122214121121212222 21
1111214112111211122114111111222121111122111111211111221222222
2114212112111112112111121411142141121222111221111214212412 21
2111112141222121141212112221121111114221141122121221224211141

#105 (mean 1.592, $|D_3|$=0.229):
1111111111111111111111111111111121212211111212211221121141122 1
1121211111111121122111222122242111214211242122211212224222 22
1124212121111112122211224411122222421222112111222124221222241
2221122411211212411122121212124211222111211211121212411222 12

#106 (mean 1.583, $|D_3|$=0.232):
1111112114222212212121112411111211222241142211221221114221 11
2111222212111221122212221121121111222212111212121212422211 22
2121221111121122111112212111122211142242112142211221124121 42
1221411141212112112221111112111112242111221211111221112211 12

#107 (mean 1.579, $|D_3|$=0.218):
2422222222222122111111111122214121222111111221122111111111111
1212111111111421211221111114222241121211112111111111121212221
2221212221111211121211422111121212114121421121111221212212211
1121111111441221221111422221214122221112422211412222211111424

#108 (mean 1.579, $|D_3|$=0.214):
2422222222222222121221211221112121122122112111122224214414122
1112111111211111411211112221111111221421111221111111142211122
1111111222222222122211414111141421112121111111121121214111212
1111221414411112222111112121112111111242112222111111211211141

#109 (mean 1.592, $|D_3|$=0.220):
1111111111111122421242142141122111214212221211221112421112144
2111212111112214114142121112124211111111112212211112421421 21
1121422122221111211111124124221111111121211211111212211211212
4121111111221111211121111111121211142422211412242122221112122

#110 (mean 1.592, $|D_3|$=0.220):
1111111111222142112122212212111111411214211124124112111421 21
2111112111111111124112121211111222222124112111114222212211 22
2422222111111211421211111111211121122442111121111121211 24211
1122122111121111112141112214211421221221242114212141121112 14

#111 (mean 1.579, $|D_3|$=0.229):
2222222121221211211112121121112111111121142122212211111141212
1241121212212111111221212221124111112222122111211211112112 22111
1224141111111122112444121221212111112112242122221121221122 21
1121212211122112121121212124112121122111212421112112214224 21

#112 (mean 1.583, $|D_3|$=0.218):
1111111111121211121141221221412211122211111211141111111122111
2411214212111121212124114221124221111211122221441412142221 21
1121211212221112222112111211211111111212211111121121111121121
2124421114111222121221122211222111124424221112111442212111 11

#113 (mean 1.579, $|D_3|$=0.248):
2222422222222222222211212221112222111211111422111212121112 41
1111122121212122421111211222111221121122121211112112211111111
1122221122111121121222112111114221142121221121414111122111 12
2124112112121111211221121122141222211214222121214222111111 12

#114 (mean 1.579, $|D_3|$=0.229):
2222222222222222221111112112112111121212121241111224111122 22
1122222122111112111111211212111411222114124111111222112411211
2112142121111111111114412112221221111212114212411212111221222
1422121211222212122211212112121211141112112112112111112214211

#115 (mean 1.592, $|D_3|$=0.220):
1111111111111111121121121114222221221222224221221121111114111
2222121211121111111111441114221212121411124221412421121121 12
1212114111212212221111112412144221112222122121114121121241211
1222114411121121211121112111111211114211112224114222112111 11

#116 (mean 1.592, $|D_3|$=0.233):
1111111111111111111111111111111111111111121121121142122212 44
1121114221241111212412221211422221212421221121212221112222 12
2214144121141111111212124211412142141212421221111112111122 22
1121111111121141211211124111112242221411212221241241111211 11

#117 (mean 1.579, $|D_3|$=0.222):
2222222211222211114111112221421122121242211221111111122241221
2112112112112221112111212111121111121112221224211111212114 22
1111111211111212222411111211211221111121112142111111221121222
4122221122121411111214122222111112121141242112412221141122 21

#118 (mean 1.587, $|D_3|$=0.214):
1241222422222141112121111111212121121221111212111211111214221 11
2112211111211111111221141111222111112222121111211121111124221 2111
1112211242111221111114112111212211112142112242211122222122 4144
1242212222111111421211111211111242411212222211111112112122121142

#119 (mean 1.579, $|D_3|$=0.218):
2222422222222222422222222222222222224211211211214111111111221142
1112121111211121212221221121111221211112211121121111111121124111
1112111122111142111111112111111124142122422141111222212111 1122
1411211221111111121121211412112211111121121111111141422241211

#120 (mean 1.579, $|D_3|$=0.214):
2222224222242222222122411212122111121111111111212112112121 2111
2122111211211112112111111211412211221112112211122224141111 2111
1211221212111122114121121211412211111411112212111111211422444
4112111111212112222411412212412411121211212111111112112121 1221